

SHAHZAD ALI

Python Backend Developer | Django · DRF · FastAPI · REST APIs · AWS

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PROFESSIONAL SUMMARY

Python Backend Developer with 2+ years of experience and 13+ production projects delivered, building scalable web applications using Django, DRF, and FastAPI. Skilled in REST API development, database optimization, asynchronous systems (Celery + Redis), and cloud infrastructure (AWS EC2, S3, RDS). Experienced in VPS deployment, Linux server management, and React API integration. Proven track record of delivering measurable performance gains in production environments.

KEY HIGHLIGHTS

- **Backend Architecture:** Proficient in designing high-availability applications using Python, Django, DRF, and Flask. Strong expertise in RESTful API design (pagination, versioning, filtering), third-party integrations, and database management.
- **Advanced Web Scraping:** Specialist in large-scale data acquisition with 87+ websites successfully scraped, handling dynamic JS-heavy pages and anti-scraping protections.
- **Full-Stack Capability:** Skilled in building responsive UIs with HTML, CSS, JavaScript (AJAX), and Bootstrap; experienced in React API integration for seamless end-to-end delivery.

SKILLS

Languages: Python, JavaScript (ES6), SQL, HTML5, CSS3

Frameworks: Django, DRF Expert, Flask, FastAPI, Celery, Redis, React (API Integration)

Backend Concepts: REST API Design, JWT/OAuth Auth, Caching, Async Processing, Query Optimization, DB Indexing, 3rd Party API Integration, FastAPI Microservices

Architecture: Django MVT, Service Layer Pattern, Clean Code, Scalable App Structure

Databases: PostgreSQL, MongoDB, SQLite, Django ORM

Cloud (AWS): AWS EC2, AWS S3, AWS RDS, AWS IAM, AWS CloudWatch, AWS Route 53

DevOps & Tools: Docker, Nginx, Gunicorn, GitHub Actions, Git, Postman, VS Code, VPS Deployment & Management, Linux Server Administration, SSL Configuration

Testing: Pytest, Django TestCase, Unit Testing

Scraping: Scrapy, Selenium, BeautifulSoup, Requests, Pandas

Soft Skills: Team Collaboration, Agile/Scrum, Problem-Solving, Code Review & Mentoring

3rd Party APIs: Twilio (WhatsApp API), Mapbox API, SendGrid, Stripe, PayPal, Mailgun, Backblaze B2

Collaboration Tools: Cliq, Microsoft Teams, Discord, Zoho Projects, Zoho Bug Tracker, Ticket System (Jira/Trello), Miro

PROFESSIONAL EXPERIENCE

WebBuggs

JAN 2025 – Present

Python Developer

Architect and implement backend APIs using Django, Flask, FastAPI, and Celery. Manage full development lifecycle from design to cloud deployment on AWS and VPS environments.

- Engineered REST APIs and core automation workflows at CRMCopilot.ai, eliminating 30 minutes of manual work weekly and enhancing data consistency across teams.
- Refactored a critical Django backend service, boosting performance by 40% through query optimization (select_related, prefetch_related) and database indexing.

- Designed and deployed a modular web scraping framework for 87+ websites using Selenium, Scrapy, and BeautifulSoup, including CompareTheBuild, Hello-Living, and Premier Snag.
- Implemented Celery + Redis for asynchronous task processing, ensuring system responsiveness under heavy load.
- Deployed and managed applications on AWS (EC2, S3, RDS) and VPS (Linux/Nginx), handling server configuration, SSL setup, and process management with Gunicorn.
- Integrated FastAPI microservices for high-performance endpoints and connected React frontends via RESTful API contracts.

BNR360

JUNE 2024 - DEC 2024

Django Developer

Backend developer focused on scalable Django applications, RESTful APIs, and PostgreSQL optimization.

- Enhanced API response times by 25% through query optimization and Redis caching strategies.
- Designed secure JWT-based authentication and permission systems using Django REST Framework.
- Collaborated with frontend (React) teams to streamline API integration, improving load times by 20%.
- Managed AWS RDS (PostgreSQL) instances and configured S3 for media/static file storage.

Website Access

JULY 2023 - MARCH 2024

Full-Stack Developer

Developed and maintained web applications using Python, Django, Bootstrap, and JavaScript for a UK-based client.

- Collaborated with UX/UI teams to design responsive interfaces, improving user engagement by 20%.
- Implemented AJAX for dynamic content updates and integrated third-party APIs (payment gateways, auth systems).
- Ensured 99.9% uptime through deployment support and ongoing maintenance.

KEY PROJECTS

WebBuggs Projects

CRMCopilot.ai - AI-Powered CRM Assistant [Python, DRF, FastAPI, OpenAI API, PostgreSQL, Celery, Redis, AWS]

- Integrated OpenAI API (GPT models) to build an NLP engine for summarizing call transcripts and generating context-aware email drafts.
- Engineered secure OAuth-based integrations with Salesforce, Zoho, and HubSpot for real-time CRM data sync.
- Implemented Celery + Redis for async email batch processing, keeping the main app responsive under load.
- Designed versioned REST APIs with DRF and FastAPI for seamless React frontend integration.
- Deployed on AWS EC2 with RDS (PostgreSQL) and S3 for file storage; managed via GitHub Actions CI/CD.

CompareTheBuild - E-Commerce Comparison Platform [Python, Flask, Scrapy, MongoDB, Celery, Redis]

- Built a modular scraping system extracting product data (names, prices, specs) from 87+ e-commerce sites.
- Developed automated price monitoring with Celery and Redis for real-time product updates.
- Created a product matching algorithm to group similar items across retailers for easy comparison.

Hello-Living - Property Hosting CRM [Django, DRF, PostgreSQL, Celery, Redis]

- Developed multi-role authentication (hosts, clients, admins) with JWT and role-based permissions.
- Built property listing and booking management with real-time reservation handling.
- Created admin panel for deal management, analytics reporting, and dispute resolution.
- Integrated automated email notifications using Celery for booking confirmations and reminders.

Premier Snag - Property Inspection Booking [Django, DRF, PostgreSQL, AWS S3, EC2, Nginx]

- Implemented role-based access control for clients, inspectors, and administrators.

- Built a booking system for scheduling inspections with automated inspector assignment logic.
- Created PDF inspection report generation with photo uploads stored on AWS S3.
- Deployed on AWS EC2 with Nginx + Gunicorn, configured SSL and domain via Route 53.
- Integrated payment processing with invoice generation and payment tracking.

Advanced Web Scraping Engine (87+ Websites) [Python, Scrapy, Selenium, BeautifulSoup, Pandas]

- Deployed custom spiders for IEEE Xplore, SpringerLink, ScienceDirect, and 87+ e-commerce platforms.
- Engineered hybrid approach: Selenium for JS-rendered content + Scrapy/BS4 for static HTML parsing.
- Implemented anti-bot evasion: proxy rotation, user-agent switching, intelligent rate limiting.
- Built resilient data pipelines with retry logic and error handling, outputting clean JSON/CSV to PostgreSQL.

BNR360 Projects

Sphere - Recruitment Management Platform [Django, DRF, PostgreSQL, JWT]

- Developed scalable RESTful APIs (with pagination, filtering, throttling) for job postings and applicant tracking.
- Optimized PostgreSQL with strategic indexing and query tuning, reducing load times by 25%.
- Implemented JWT + Django Allauth for secure authentication and authorization.
- Integrated analytics and reporting features for real-time HR decision-making.

Website Access Projects

ClickHub - Full-Featured E-Commerce Platform [Django, PostgreSQL, AJAX, Bootstrap]

- Designed robust e-commerce platform with relational database management and optimized queries.
- Implemented AJAX and Bootstrap for dynamic, user-friendly interfaces.
- Delivered from inception to deployment; achieved A grade and 6/6 GPA.

EDUCATION

Bachelor of Science in Computer Science

University of South Asia - Lahore, Pakistan

2020 - 2024

Intermediate in Computer Science (ICS)

Punjab College Raiwind - Lahore, Pakistan

2018 - 2020